

Executive decision

- **Aligned NV found.** Candidate image145844_reimage_r03 tracked at [117.314436, 117.761644, 115.141679] um with 43.535 kcps, passed the human-requested strong-pi pulsed-ODMR alignment screen, and remained usable for targeted follow-up.
- **Resonance calibration.** Strong-pi pODMR supports a clear resonance at the 3.875 GHz grid point. The latest weak-pi pODMR refresh supports a grid center at 3.8765 GHz, with several-100-kHz grid-level uncertainty only.
- **T2* conclusion.** No claim-grade numeric T2* was established for r03. Five Ramsey datasets were analyzable and healthy, including a final 1.0e6-shot-per-tau refreshed-center run, but the programmed carrier was not a unique stable model. The best fixed-carrier fits give order-1–3 us descriptive candidates only.
- **13C conclusion.** The Ramsey/FFT/least-squares branch supports a negative/unsupported nearby-13C conclusion under these conditions: expected sidebands at det minus/plus the 13C Larmor offset were not a coherent pair, and the explicit carrier-plus-sidebands model was not favored.
- **Operational recommendation.** Close the r03 branch and do not run more automatic Ramsey repeats. If a numeric T2* value is mandatory, that should be a new protocol-development question (for example phase-stability/early-time Ramsey redesign) rather than another bridge job in this project.

Evidence basis

Step	Decision-relevant evidence
Image and tracking	Original saved image axis contract was verified as ImageData_YXZ. After r01 count/focus collapse, a fresh Imaging scan over the same region used an explicit bridge-recorded Y-X-Z permutation. r03 was the first candidate with clear strong-pi pODMR resonance after r01/r02 were rejected.
Strong-pi pODMR	3.825–3.925 GHz, 11 points, 4 x 20000 repetitions, mod_depth=1, length_rabi_pulse=52 ns. Raw signal, point-wise ratio, and reference-line normalization all had deepest dip at 3.875 GHz; raw signal drop was about 16.6% vs edge median; all four averages supported the center dip; no drift flags.
Weak-pi pODMR refresh	3.8745–3.8775 GHz, 31 points, 4 x 50000 repetitions, mod_depth=0.1, length_rabi_pulse=0.57 us. Raw signal and fitted-reference-line normalization minimized at 3.8765 GHz; raw drop about 15.3%; no drift flags.
Final Ramsey	auto__ramsey, mw_freq=3.8765 GHz, det=1.5 MHz, tau=48 ns..8.048 us in 41 points, 20 x 50000 repetitions. Completed normally with final_counts=43.433 kcps, no monitor error, no stop request, and no scan-order-aware drift flags.
Model comparison	In the final Ramsey, the fixed 1.5 MHz carrier-like component was present but modest: ratio amplitude 0.01575 full span / 0.01231 after skipping first four tau points; raw signal LS amplitude 0.705 kcps, below the median per-point signal SEM 0.850 kcps. BIC preferred a damped high-edge 2.266 MHz component over fixed carrier by about 10 and over carrier-plus-expected-13C-sidebands by about 28.

Ramsey branch summary

Run	Shots / tau	Outcome
Initial det=1.5 MHz scout, 0–6 us	2.0e5	Analyzable, no drift flags, but strongest exploratory component near 0.884 MHz rather than programmed carrier or expected 13C sidebands. No claim.
Det=1.0 MHz 8-us follow-up	4.0e5	Programmed carrier weak; largest combined component near 1.178 MHz; per-average screens inconsistent. No claim.
Short-tau high-SNR det=1.0 MHz	1.08e6	Persistent empirical near-1.192 MHz component and descriptive T2* about 6.33 us, but not tied to the programmed carrier/sideband model. No claim.
Short-tau det=1.5 MHz shift check	1.08e6	Useful evidence against simply promoting the fixed 1.192 MHz feature, but skip-transient and per-average views remained mixed. No claim.
Refreshed-center det=1.5 MHz 8-us final run	1.0e6	Healthy terminal run. Carrier-like signal present but not model-unique; high-edge component and model-choice dependence dominate. No numeric T2* claim; no supported 13C sideband/coupling evidence.

Why more automatic hardware is not justified

The final Ramsey already reached 1.0e6 shots per tau point with stable bridge execution and no drift flags. The fixed-carrier raw LS amplitude was 0.705 kcps, below the measured median per-point signal SEM of 0.850 kcps. If this were only shot noise, reaching SEM 0.25 kcps would require roughly $(0.850/0.25)^2 = 12$ times more shots per point; a conservative 5-sigma view of the 0.705 kcps carrier would require about $(0.850/0.141)^2 = 36$ times more shots per point. That is not a good autonomous use of the bridge because the limiting problem is not just shot noise: the preferred model is a high-edge component and the expected 13C sidebands do not form a coherent pair.

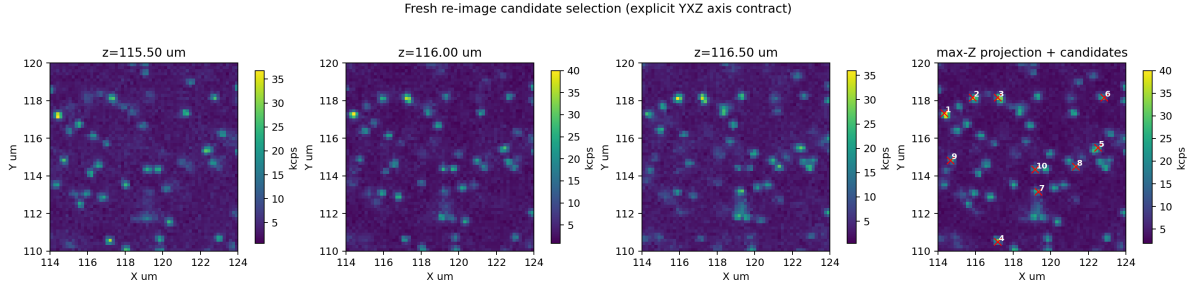
Therefore a further long-span or higher-shot repeat would mainly accumulate evidence for the same ambiguity. A new T2* attempt would need a changed scientific protocol and a new explicit model, not a repeat of the current Ramsey branch.

Literature and prior-result comparison

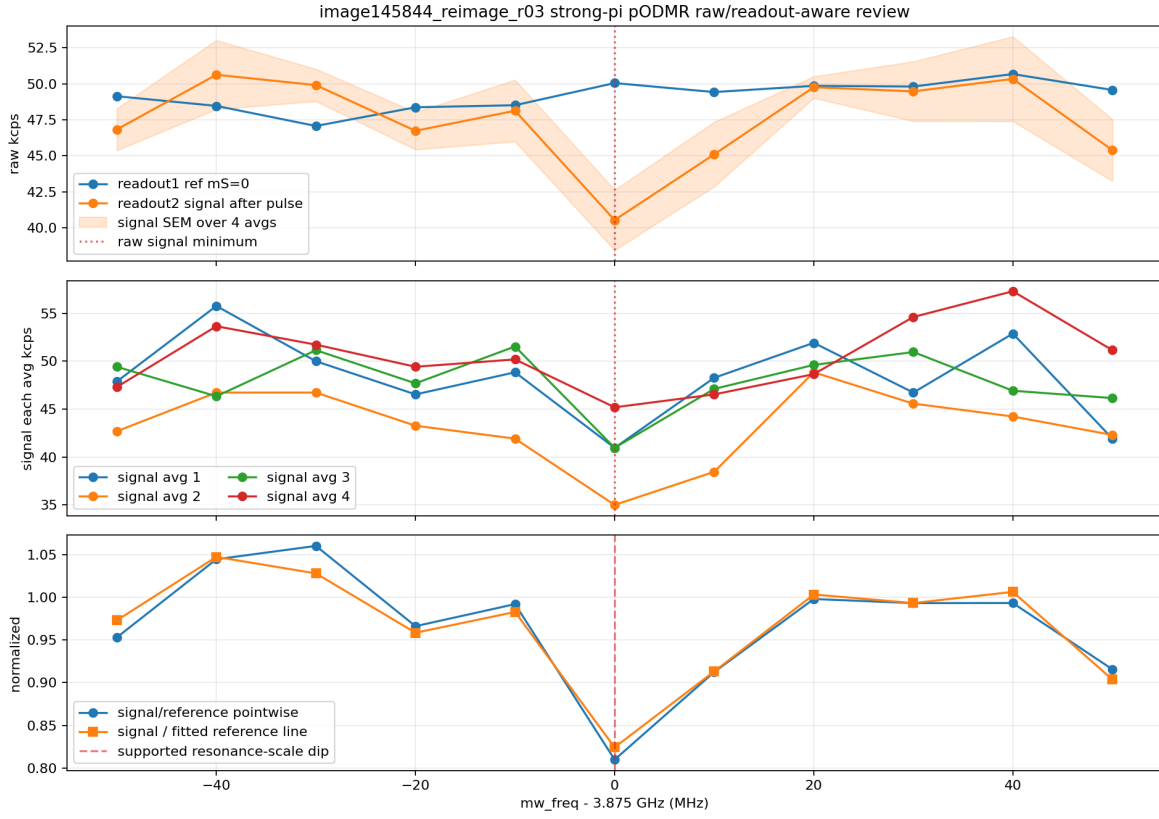
Local literature index checked: Doherty et al., Physics Reports 528, 1–45 (2013), doi:10.1016/j.physrep.2013.02.001, and Degen/Reinhard/Cappellaro, Rev. Mod. Phys. 92, 015004 (2020), are the relevant local review references for NV center Ramsey/free-induction practice and quantum-sensing interpretation. DOI web-fetch attempts for those two references returned no acceptable resource in this environment, so the local library/index and project evidence were used. Lab prior-result guidance in NV_RESEARCH_KNOWLEDGE also explicitly warns that mixed-spectrum high-shot Ramsey data should not be promoted to numeric T2* without signal uniqueness and model stability.

Decision-relevant comparison: a Ramsey T2* or nearby-13C claim should rest on a signal tied to the intended carrier/physical sidebands and robust against baseline/model choice. The r03 branch does not meet that bar. The negative 13C statement is therefore scoped to “no supported nearby-13C Ramsey sideband/coupling evidence under these measurement conditions”, not to an absolute absence of every possible weak nuclear spin.

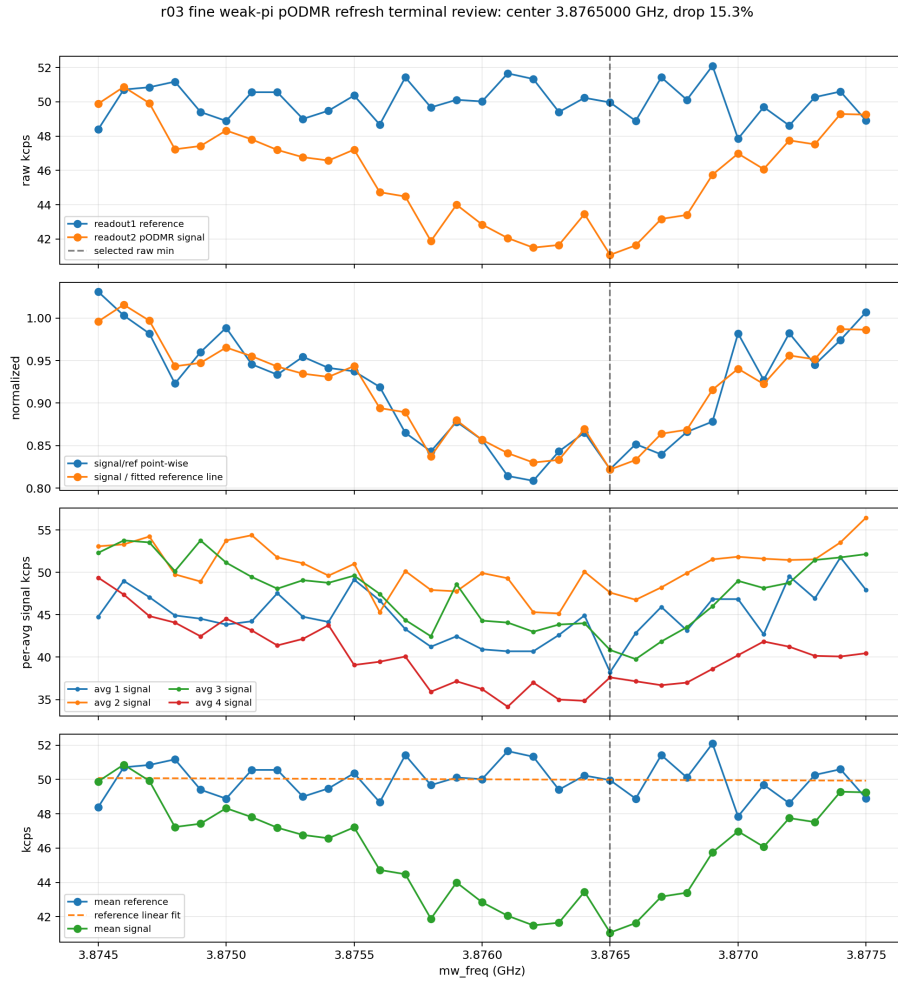
Figures



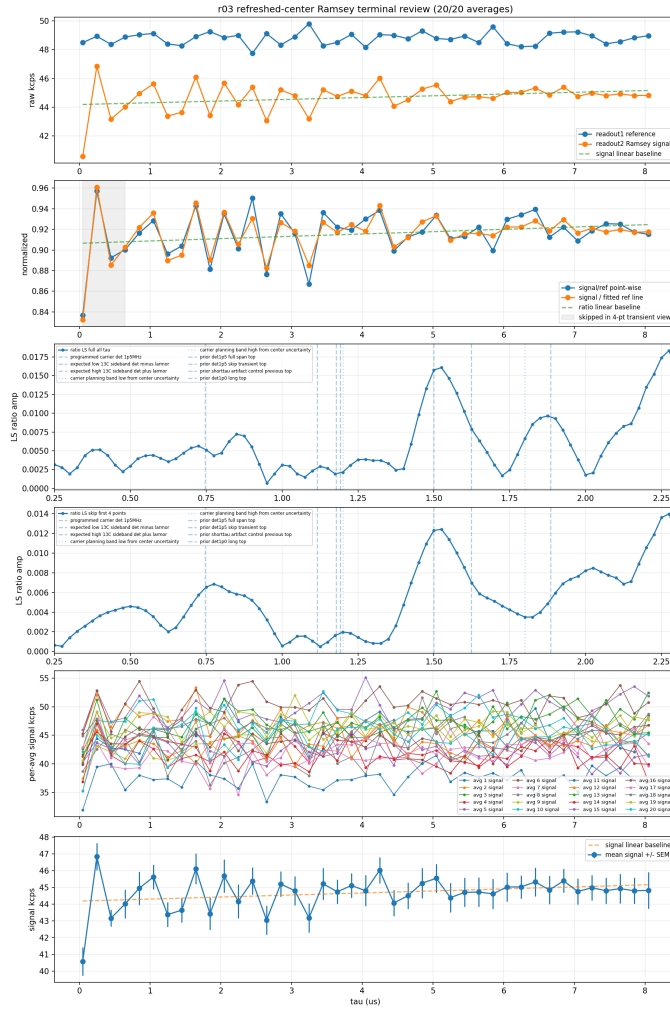
Fresh re-image candidate selection. r03 was followed after r01 and r02 failed the strong-pi pODMR resonance screen.



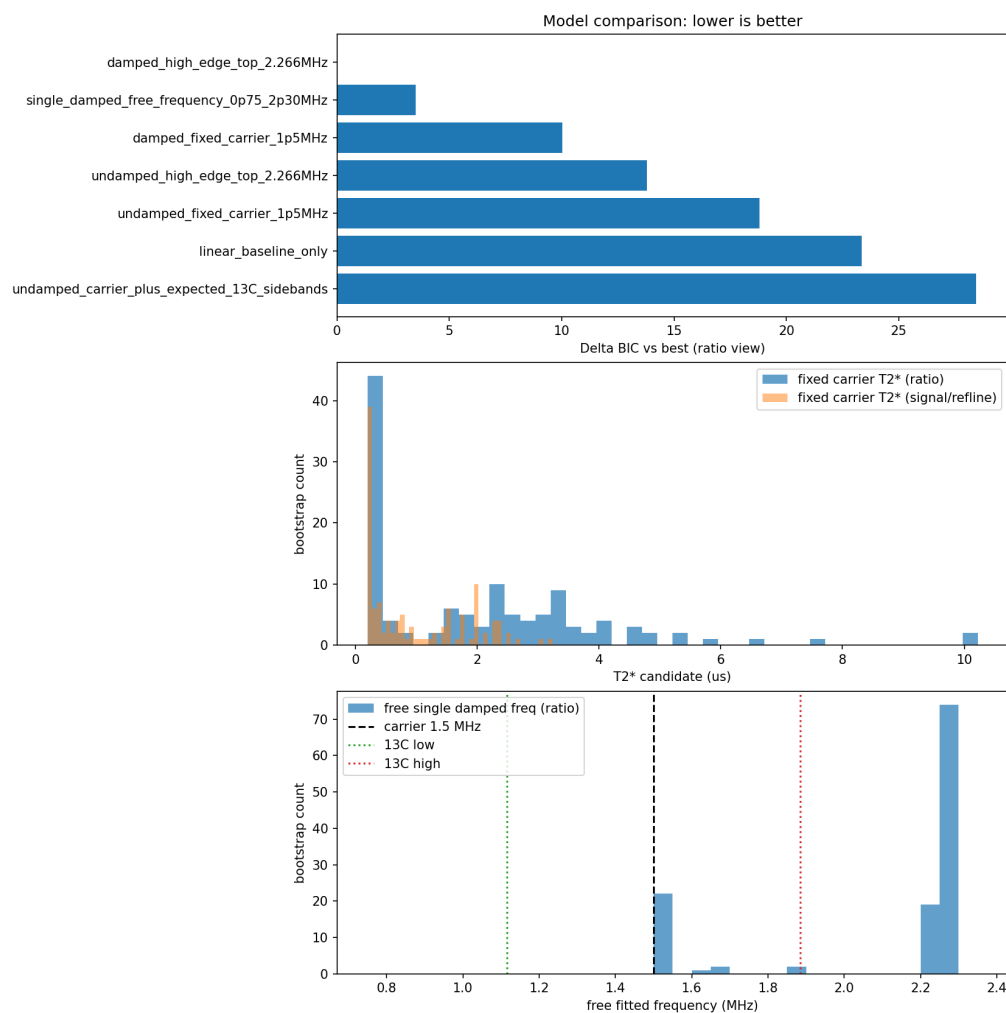
r03 strong-pi pulsed ODMR. The 3.875 GHz grid point is the clear raw/readout-aware resonance used to accept r03 as aligned.



Final weak-pi pODMR refresh. The grid-supported calibration center is 3.8765 GHz for the final Ramsey branch.



Terminal refreshed-center Ramsey review. The run was healthy and high-shot, but the carrier/sideband interpretation was not unique.



Model comparison. Lower BIC is better; high-edge models beat the fixed-carrier and carrier-plus-expected-13C-sideband hypotheses.

Final project-scoped claims

1. image145844_reimage_r03 is a magnetic-field-aligned NV candidate suitable for targeted follow-up.
2. The current project did not establish a claim-grade numeric T_2^* for r03. Descriptive fixed-carrier fits are order 1–3 μs , but are not robust enough to quote as the result.
3. Under the completed Ramsey/FFT/least-squares measurements, there is no supported nearby- ^{13}C side-band/coupling evidence for r03.